

Skel-WAM: A Hand-Skeleton-Conditioned World Action Model for Human-to-Robot Manipulation Transfer

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Project page: <https://HealorCai.github.io/Skel-WAM>

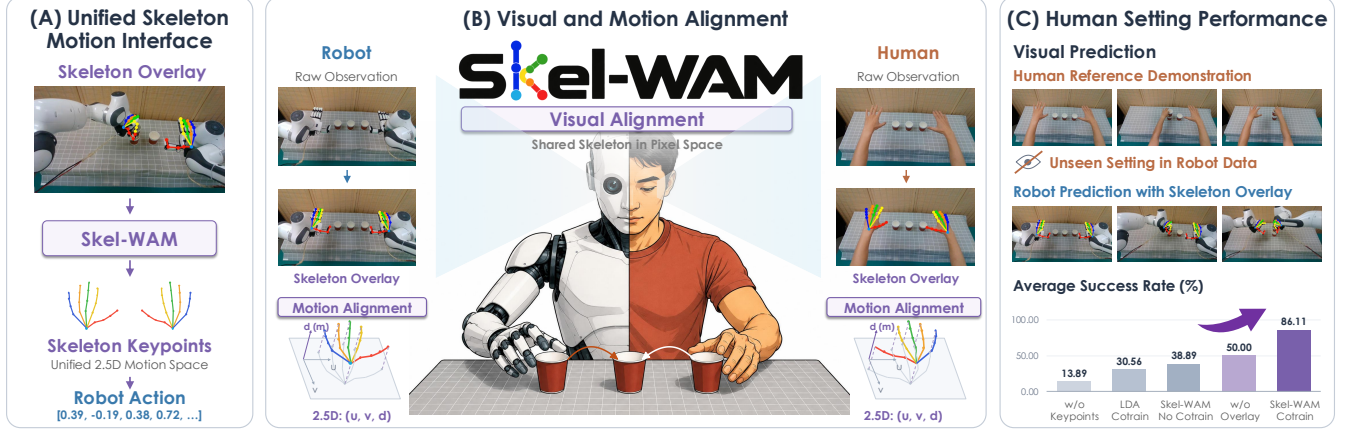


Fig. 1. **Skel-WAM** is a world action model that bridges human and robot manipulation through a unified hand-skeleton motion interface. Scene-grounded skeleton overlays and structured 2.5-D keypoints represent human and robot motion in a shared space, enabling cross-embodiment alignment and joint prediction of visual dynamics and skeletal motion. This representation allows Skel-WAM to learn robot-missing task variations from human demonstrations without robot action labels, raising average success from 38.89% to 86.11%.

Abstract—Robot demonstrations are expensive to collect and often provide limited distributional coverage of task variations. Human videos offer a low-cost source of complementary manipulation experience, but learning from them requires bridging embodiment gaps in visual appearance and action spaces. We introduce Skel-WAM, a world action model that bridges these differences through a unified hand-skeleton motion interface. The key insight is to align human and robot motion through a common hand topology, combining skeleton overlays that ground motion in the scene with structured 2.5-D keypoints that encode explicit hand kinematics. Video and Keypoint Experts jointly learn visual and skeletal dynamics through a Mixture-of-Transformers, while a separate robot-trained Action Expert maps these predictions to executable controls. This separation enables human and robot demonstrations to directly supervise shared dynamics without requiring robot action labels for human videos. Across four real-world bimanual tasks and seven simulated tasks, Skel-WAM achieves average success rates of 79.86% and 63.29%, surpassing the strongest baseline by 22.22 and 8.28 percentage points, respectively. Human-robot cotraining more than doubles real-world success on task variations absent from robot training data, from 38.89% to 86.11%. These results demonstrate that a shared skeletal interface enables joint learning across human and robot data and expands robot task coverage through complementary human demonstrations.

I. INTRODUCTION

Scaling robot demonstration datasets has accelerated progress in robot manipulation, but robot data collection remains costly and time-consuming. Moreover, increasing

dataset size does not guarantee coverage of task variations. For a fixed instruction, demonstrations may capture a nominal task while remaining sparse across object position, required rotation, environmental disturbance, or manipulation height. A policy can therefore fit the training distribution and still fail on nearby, task-consistent cases. Scaling robot datasets improves generalization, but collecting every missing condition for a particular task and workspace remains costly.

Human videos provide an abundant source of complementary experience. They can be collected at scale and naturally contain diverse hand-object interactions. Prior work has used them for video-based planning [1] and kinematic retargeting [2]. However, converting this diversity into robot control requires bridging both appearance and action gaps. Human and robot hands differ in morphology, and human videos lack the joint positions, end-effector poses, or actuator commands used to supervise robot policies. Human-specific models such as MANO [3] provide kinematic structure but not robot actions. Dense visual motion lacks finger-level semantics, while direct retargeting relies on embodiment-specific correspondences.

Alignment alone also leaves the role of human data underspecified. Treating humans as another embodiment enables joint learning, but leaves their concrete contribution to robot policies unclear. We investigate whether human demonstra-

tions can expand within-task coverage by supplying variations absent from robot data under the same instruction. This requires an interface that aligns motion across embodiments, captures dexterous hand structure, and grounds that motion in the scene.

We introduce Skel-WAM, a world action model with a unified hand-skeleton motion interface that maps human and robot hands to a common two-hand topology with 21 keypoints per hand. Complementary skeleton overlays ground motion in the scene, while structured 2.5-D keypoints preserve finger identity and explicit trajectories. Video and Keypoint Experts form a Mixture-of-Transformers to learn shared visual and skeletal dynamics from human and robot data. A separate robot-trained Action Expert decodes these predictions into embodiment-specific controls, allowing human videos to supervise shared dynamics without corresponding robot action labels. We evaluate Skel-WAM on four real-world bimanual tasks and seven simulated manipulation tasks. It achieves the best average performance among all evaluated methods. We further hold task language fixed and use human demonstrations to cover robot-missing variations in position, rotation, disturbance, and height. Co-training raises average success on these cases from 38.89% to 86.11%, a 47.22-point gain. Furthermore, controlled ablations confirm the complementary roles of overlays and keypoints.

Our main contributions are threefold:

- a unified hand-skeleton motion interface that aligns human and robot motion through visual overlays and structured 2.5-D keypoints,
- a decoupled Video-Keypoint MoT and robot-only Action Expert that learn from human videos without robot action labels, and
- extensive evaluations in the real world and simulation that show consistent gains over strong baselines and a 47.22-point improvement from human-covered task variations.

II. RELATED WORK

A. Human Videos for Robot Learning

Prior work uses human videos for video or trajectory planning [1], [4]–[6], retargets estimated human motion into robot actions [2], [7], [8], or synthesizes robot-format observations by replacing visible human hands and arms with robot embodiments [9]–[12]. HandEdit benchmarks dexterous human-to-robot image editing [13]. Another line of work treats humans as an additional embodiment during joint training [14]–[18]. Other approaches learn from structured human recordings [19], [20] or use human videos as in-context task specifications [21], [22]. Skel-WAM instead studies how human demonstrations can expand within-task coverage by supplying robot-missing task variations under fixed task instructions.

B. Cross-Embodiment Motion Representations

Effective transfer requires a motion representation that remains comparable across visual and kinematic differences.

Prior work uses latent actions, flow, masks, and sparse tracks or affordances [5], [6], [23]–[28]. MANO and retargeting provide explicit human kinematics [2], [3]. More closely related methods explicitly model hand structure. DexWM uses MANO keypoint displacements [29], Ego-Pi overlays finger-identified skeletons while mapping human joints into robot action space [17], and OSCAR conditions an omnibodiment world model on a common 2D skeleton [30]. Skel-WAM instead models human and robot motion in a shared two-hand skeleton space and jointly predicts future video and trajectories over 21 keypoints per hand.

C. World Action Models for Manipulation

A shared representation must also connect predicted futures to control. ACT and Diffusion Policy predict action chunks directly [31], [32], whereas VLAs scale observation-to-action learning [33], [34]. Video policies recover control from predicted futures [1], [35]–[37], and generated videos yield plans through reconstruction, flow, or inverse dynamics [38]–[40]. World action models (WAMs) unify these views by coupling dynamics and control through causal, diffusion, or latent modeling [41]–[45]. Recent variants add masks and 3D targets [25], [46], [47], factorized or updatable simulation [48], [49], and human-video pretraining [50]. Skel-WAM jointly predicts video and skeleton with a Video-Keypoint MoT, then decodes control through a robot-only action expert. Human demonstrations therefore supervise shared world and motion dynamics without requiring fabricated robot actions.

III. METHOD

In this section, we describe Skel-WAM in detail. We begin by defining the learning setting with action-labeled robot demonstrations and human demonstrations without robot action labels (Sec. III-A). We then introduce the unified skeleton motion interface, comprising scene-grounded overlays and structured 2.5-D keypoints (Sec. III-B). Next, we present the Video-Keypoint Mixture-of-Transformers for learning shared visual and skeletal dynamics across human and robot demonstrations (Sec. III-C). Finally, we describe human-robot cotraining of the shared model, robot-only training of the causal Action Expert (Sec. III-D), and the flow-matching objectives and implementation details (Sec. III-E).

A. Problem Formulation

At time t , the inputs are a task instruction ℓ , the current RGB image i_t , and sampled skeleton keypoint history $K_t^{\text{hist}} = \{k_{t-H_h}, \dots, k_t\}$. Let v_t denote the frozen video VAE latent of the current skeleton-overlay image $\tilde{i}_t$. The model predicts an action chunk $A_t = \{a_t, \dots, a_{t+H_a}\}$ through intermediate overlay-video latents $V_t = \{v_t, \dots, v_{t+H_v}\}$ and structured keypoints $K_t = \{k_t, \dots, k_{t+H_k}\}$, both anchored at the current observation.

Human and robot demonstrations share a skeleton topology but differ in action supervision: $\mathcal{D}_R =$

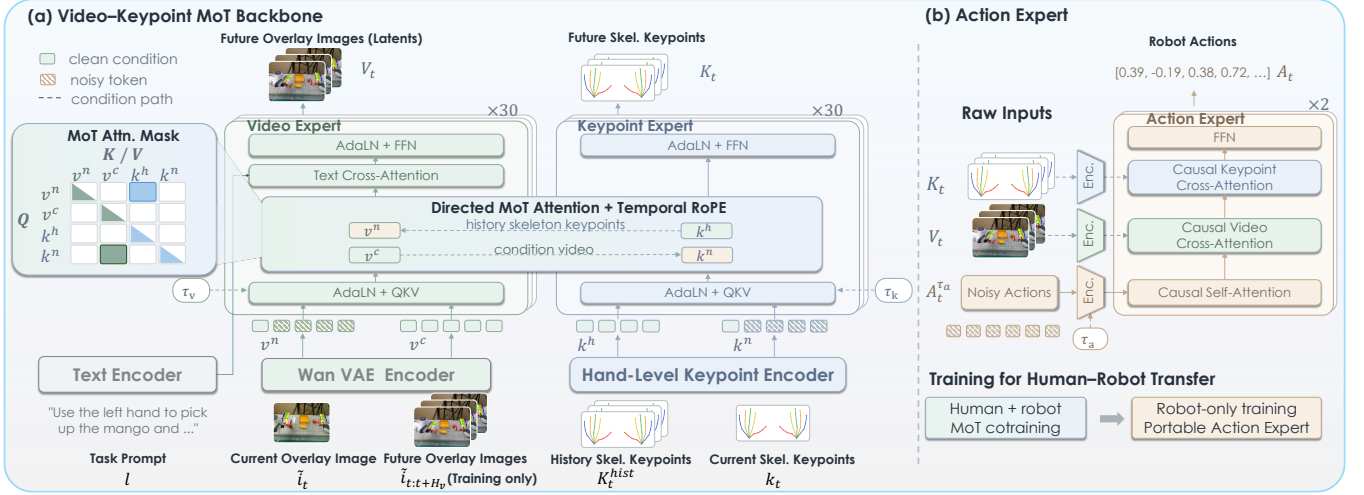


Fig. 2. **Skel-WAM architecture and cotraining.** (a) The Video-Keypoint MoT jointly predicts future overlay-video latents and structured 2.5-D keypoint trajectories from human and robot demonstrations. (b) The causal Action Expert maps these predictions to robot actions. Human and robot videos cotrain the MoT, while only robot data train the Action Expert, enabling learning from human videos without robot action labels.

$\{(I^R, K^R, \ell, A^R)\}$ and $\mathcal{D}_H = \{(I^H, K^H, \ell)\}$. We factor shared prediction as

$$\begin{aligned} p_\theta(V_t, K_t \mid v_t, K_t^{\text{hist}}, \ell) \\ = p_\theta^V(V_t \mid v_t, K_t^{\text{hist}}, \ell) p_\theta^K(K_t \mid V_t), \end{aligned} \quad (1)$$

and decode robot actions through $p_\phi(A_t \mid V_t, K_t)$. Both datasets supervise p_θ , whereas only $\mathcal{D}_R$ supervises p_ϕ , separating shared dynamics from robot-specific control.

B. Unified Skeleton Motion Interface

We represent human and robot hands with a shared topology of two hands and 21 keypoints per hand. Each skeleton is expressed in two synchronized forms: a scene-grounded visual overlay and structured 2.5-D keypoints. Together, they preserve hand location, finger configuration, and bimanual motion without requiring correspondence between human and robot joint commands.

Visual skeleton overlay. Skel-WAM uses skeleton-overlay images for both observed conditions and future video prediction. During training, overlays are rendered on observed and future RGB images. At inference, the Video Expert directly generates future overlay images. For human video, WiLoR [51] estimates MANO parameters and image-space keypoints. We recover metric keypoints using the measured hand size and a two-stage camera-space optimization that first estimates wrist translation and then jointly refines translation and rotation. For robot videos, we select semantically corresponding URDF links, compute their positions by forward kinematics, and transform them into the camera frame using calibrated extrinsics. Camera intrinsics project both human and robot keypoints into image coordinates while retaining metric depth. We construct $\tilde{i}_t = \text{Overlay}(i_t, k_t)$ by drawing valid keypoints and connections on each RGB image (Fig. 3). Visible elements are rendered in depth order with depth-adaptive marker sizes, while valid out-of-frame points remain only in the structured representation. The overlays



Fig. 3. **Skeleton overlays across embodiments.** (a) Inspire dexterous hands in simulation. (b) Human hands and (c) RobotEra XHand1 dexterous hands in the real world.

thus provide a shared, scene-grounded representation of human and robot hand motion.

Structured 2.5-D keypoints. The same skeleton is retained as structured 2.5-D keypoints, providing an explicit, low-dimensional representation of hand motion. Each keypoint contains normalized image coordinates $u = x/W$ and $v = y/H$, together with camera-space depth d in meters. We standardize these coordinates using fixed statistics shared across human and robot data, without clipping image coordinates. Invalid coordinates are zero-filled after normalization, and validity is encoded as $m = -0.5$ or $+0.5$. Thus, $k_{t,h}$ contains the coordinates and validity of the 21 keypoints for hand h at time t . Within the Keypoint Expert, the Hand-Level Keypoint Encoder maps the 21 keypoints of each hand at every frame to a single token:

$$k_{t,h}^r = P_k(k_{t,h}) + e_h + e_r, \quad (2)$$

where P_k is a learnable linear projection and r identifies the history condition or noise role. Temporal RoPE is applied to encode frame positions.

C. Video-Keypoint Mixture-of-Transformers

We initialize the Video Expert from Wan2.2-TI2V-5B [52]. A frozen Wan VAE encodes skeleton-overlay clips, and a shared patch projection P_v maps condition and noisy prediction latents into separate streams. Precomputed T5 embeddings condition the Video Expert through text cross-attention. Following Fast-WAM [41], we initialize transfer-

able Keypoint-Expert weights by interpolating Video-Expert parameters. Both experts contain 30 layer-aligned blocks but retain modality-specific parameters.

The experts separately embed their flow times τ_v and τ_k to generate scale, shift, and residual-gate parameters for every attention and feed-forward branch. The MoT operates on four token streams shown in Fig. 2(a): video-prediction tokens v^n , video-condition tokens v^c , observed keypoint-history tokens k^h , and keypoint-prediction tokens k^n . The current anchors in v^n and k^n remain clean, while their future tokens receive flow-matching noise. During training, future v^c tokens encode ground-truth overlay images. At inference, they encode overlays generated by the Video Expert. The video streams share P_v and use distinct role embeddings, while the skeleton streams share P_k . Each MoT block applies a stream-wise attention mask that allows each query stream to access the following key/value streams:

$$\begin{aligned} v^n &\rightarrow \{v^n, k^h\}, & v^c &\rightarrow v^c, \\ k^n &\rightarrow \{k^n, v^c\}, & k^h &\rightarrow k^h. \end{aligned} \quad (3)$$

Within-stream attention is temporally causal, while the two cross-modal paths provide full access to their condition streams. Specifically, v^n attends to the skeleton keypoint history k^h , and k^n attends to the video-condition stream v^c . Video and keypoint streams use spatio-temporal and temporal RoPE, respectively, with aligned frame positions. Mixed attention is followed by modality-specific feed-forward layers; only the Video Expert uses text cross-attention. The MoT jointly predicts video-latent and keypoint flow velocities.

D. Action Prediction and Human-Robot Cotraining

The Action Expert in Fig. 2(b) uses separate encoders to map noisy actions A_t^a , conditioning video latents V_t , and conditioning skeleton keypoints K_t into modality-specific tokens. During training, V_t and K_t are derived from ground-truth robot video and skeleton sequences, while at inference they are provided by the MoT predictions. Each of its two Transformer blocks sequentially applies causal action self-attention, causal video cross-attention, causal keypoint cross-attention, and a feed-forward layer. The causal masks allow each action query to attend only to action and condition tokens at or before its timestep. A final prediction head outputs the action flow velocity.

Human-robot cotraining first optimizes the MoT using overlay videos and skeleton keypoints from both domains as prediction targets. We then train the portable robot-specific Action Expert exclusively on robot data. Its video projection P_v is copied from the cotrained MoT and kept frozen, and action gradients do not propagate into the backbone.

E. Flow Matching and Implementation

We apply conditional flow matching [53] to video latents, skeleton keypoints, and robot actions. For each clean target $x \in \{V_t, K_t, A_t\}$, we independently sample $\epsilon_x \sim \mathcal{N}(0, I)$ and a flow time τ_x , and construct

$$x_{\tau_x} = (1 - \tau_x)x + \tau_x\epsilon_x, \quad u_x^* = \epsilon_x - x. \quad (4)$$

For V_t and K_t , only future elements receive flow noise; current anchors remain fixed. Each velocity predictor minimizes

$$\mathcal{L}_x = \mathbb{E} \left[\omega_x(\tau_x) \frac{\sum_i M_{x,i}(\hat{u}_{x,i} - u_{x,i}^*)^2}{\max(1, \sum_i M_{x,i})} \right], \quad (5)$$

where ω_x is the scheduler weight. The masks ignore invalid keypoint coordinates and invalid action steps, while retaining validity supervision for all future keypoints. Robot-only joint training minimizes $\mathcal{L}_V + 2\mathcal{L}_K + 2\mathcal{L}_a$. Human-robot MoT cotraining minimizes $\mathcal{L}_V + 2\mathcal{L}_K$, followed by Action Expert training with $\mathcal{L}_a$.

We use 1000 flow timesteps with SNR shifts of 5 for video and 1 for keypoints and actions. With probability 0.5, video-condition latents receive noise from $[0, 0.2]$, while the current anchor remains clean [42]. Each clip pairs 13 keypoint frames with 384×384 images at real-robot and simulation strides of 6 and 3, yielding 73- and 37-step action chunks. The keypoint history contains 3 and 4 frames for real-robot and simulation data, respectively. Real-robot control runs at 30 Hz with 42-D actions comprising per-arm 3-D end-effector position, 6-D rotation, and 12 hand joint angles. Simulation uses 38-D joint-angle actions. Using eight NVIDIA A800 GPUs, robot-only training jointly optimizes all experts for 80k steps. Human-robot transfer cotrains the MoT for 80k steps at a 2:1 robot-to-human ratio, then trains the Action Expert on robot data for 50k steps. Inference sequentially samples V_t , K_t , and A_t using 3/3/5 steps in the real world and 10/10/5 in simulation. All flows run from noise to data with fixed observed anchors, requiring no future observations.

IV. REAL-WORLD EXPERIMENTS

We evaluate Skel-WAM on four real-world bimanual tasks using robot and human demonstrations with complementary task coverage. We examine four aspects: (1) long-horizon performance under robot-demonstrated settings, (2) transfer of robot-missing task variations from human demonstrations, (3) contributions of skeleton overlays and structured 2.5-D keypoints to human-to-robot transfer, and (4) generalization to unseen backgrounds.

A. Real-World Experimental Setup

Benchmark. Our platform consists of two Franka Research 3 arms equipped with RobotEra XHand1 dexterous hands and an Intel RealSense D435 ego-view camera (Fig. 4(a)). We evaluate four bimanual tasks (Fig. 4(b)): **Stack Cups** sequentially stacks two side cups onto the center cup. **Pour Water** coordinates a bottle and a cup to perform pouring. **Place Fruit into Box** places a mango and an apple into a box. **Put a Wrist Guard into a Drawer** places the wrist guard into a drawer and then closes it.

Demonstrations. We teleoperate the robot end effectors using relative poses from Vive trackers and control the dexterous hands using MANUS gloves. Per task, we collect an average of 50 robot and 25 human demonstrations. Human data cover settings absent from robot demonstrations, varying

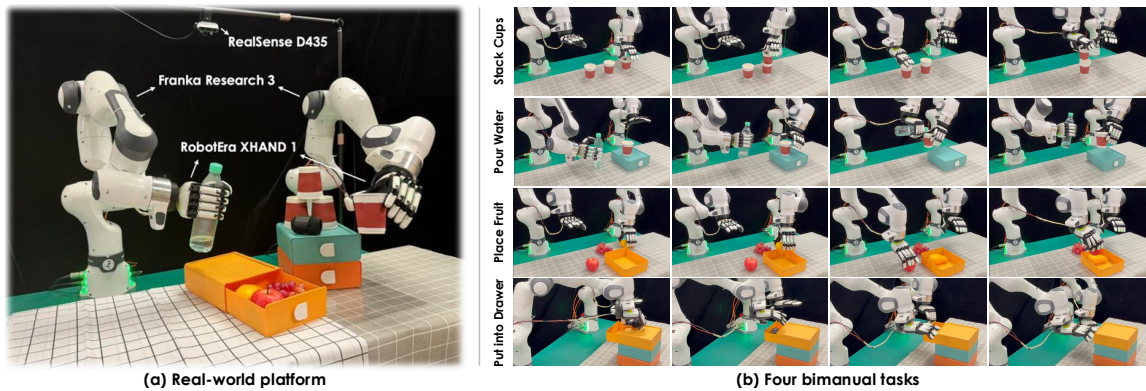


Fig. 4. **Real-world benchmark.** (a) Real-world experimental platform with two Franka Research 3 arms, RobotEra XHand1 dexterous hands, and an Intel RealSense D435 camera. (b) Execution sequences for Stack Cups, Pour Water, Place Fruit into Box, and Put a Wrist Guard into a Drawer.

TABLE I: **Real-world results with robot-only training data.** Each entry is SR / PSR (%).

Method	Avg. $\uparrow$	Stack Cups	Pour Water	Place Fruit	Put into Drawer
ACT	12.32 / 44.39	11.76 / 58.82	16.67 / 60.42	0.00 / 37.50	20.83 / 20.83
Fast-WAM	38.89 / 50.69	38.89 / 48.61	41.67 / 60.42	37.50 / 52.08	37.50 / 41.67
LDA	42.01 / 53.30	38.89 / 56.94	41.67 / 54.17	45.83 / 58.33	41.67 / 43.75
EgoVLA	57.64 / 69.10	55.56 / 76.39	70.83 / 85.42	45.83 / 56.25	58.33 / 58.33
Skel-WAM	79.86 / 89.24	77.78 / 90.28	87.50 / 95.83	70.83 / 83.33	83.33 / 87.50

object position, wrist rotation, distractors, and manipulation height (Fig. 5).

Baselines. We compare with four representative baselines. ACT [31] directly predicts action chunks from RGB observations, while Fast-WAM jointly models future videos and robot actions. LDA [43] unifies policy learning, dynamics modeling, and visual forecasting in the DINO [54] latent space. EgoVLA [7] is a unified VLA model jointly learning from egocentric human videos and robot demonstrations.

Metrics. We evaluate the Robot, Human, and New Background settings using 24, 9, and 6 trials per task, respectively. Stack Cups uses 18 Robot-setting trials to balance the setting distribution. Success Rate (SR) measures full task completion, while Progress Success Rate (PSR) measures task-specific milestones. These milestones are picking and stacking each side cup for Stack Cups, lifting the bottle and cup for Pour Water, placing each fruit for Place Fruit into Box, and inserting the wrist guard and closing the drawer for Put a Wrist Guard into a Drawer.

B. Long-Horizon Performance under the Robot Setting

Skel-WAM achieves 79.86% average SR and 89.24% PSR, exceeding EgoVLA by 22.22 and 20.14 percentage points, respectively (Table I). Its advantage extends across all four tasks, which require sequential grasping, bimanual coordination, distractor-robust localization, and precise placement. Compared with ACT, which directly predicts actions from RGB observations, Skel-WAM uses 2.5-D skeleton keypoints to provide explicit image-space position and depth. This structured motion interface reduces the grasp drift observed for ACT in Stack Cups. Fast-WAM predicts visual dynamics but lacks an explicit skeletal motion representation, leading to mistimed grasps in Stack Cups and inaccurate cup localization in Pour Water. Skel-WAM instead combines future

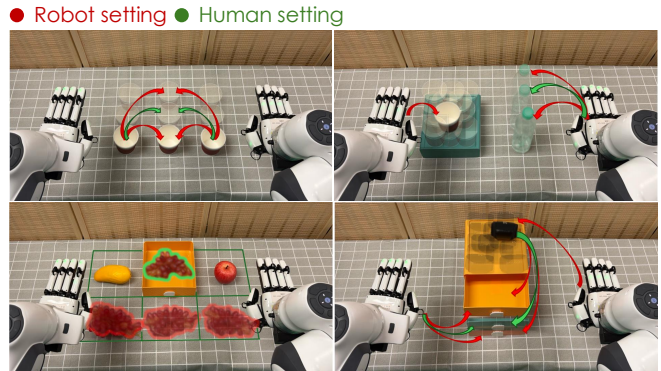


Fig. 5. **Robot and human demonstration settings across the four tasks.** Red and green denote robot and human settings, respectively. Human demonstrations introduce variations in object position, wrist rotation, distractors, and manipulation height.

visual dynamics with structured hand trajectories, providing more precise geometric and temporal guidance. LDA predicts actions through future DINO features. In Place Fruit into Box, nearby grapes bias its predicted apple-grasp location, indicating sensitivity to visual distractors. EgoVLA is the strongest baseline, but its remaining failures also concentrate on grasp and placement localization. By explicitly grounding hand motion in image position, depth, and predicted visual context, Skel-WAM achieves more reliable spatial grounding throughout long-horizon manipulation.

C. Learning Robot-Missing Task Variations from Human Demonstrations

We evaluate four robot-missing task variations covered only by human demonstrations under fixed task instructions (Fig. 5). They vary middle-cup position in Stack Cups, wrist rotation in Pour Water, distractors in the target box in Place

TABLE II: **Results on robot-missing task variations.** Cotrained models use robot and human demonstrations, whereas the no-cotrain model uses robot demonstrations only. Each entry is SR / PSR (%).

Method	Avg. $\uparrow$	Position	Rotation	Disturbance	Height
LDA-cotrain	30.56 / 49.31	33.33 / 41.67	22.22 / 44.44	44.44 / 61.11	22.22 / 50.00
Skel-WAM, no cotrain	38.89 / 61.11	11.11 / 50.00	77.78 / 83.33	66.67 / 66.67	0.00 / 44.44
Skel-WAM, cotrain	86.11 / 95.14	66.67 / 91.67	100.00 / 100.00	77.78 / 88.89	100.00 / 100.00
Cotrain w/o both	16.67 / 31.25	11.11 / 19.44	33.33 / 61.11	22.22 / 33.33	0.00 / 11.11
Cotrain w/o keypoints	13.89 / 34.03	11.11 / 25.00	22.22 / 55.56	22.22 / 38.89	0.00 / 16.67
Cotrain w/o overlay	50.00 / 68.75	44.44 / 86.11	77.78 / 83.33	44.44 / 55.56	33.33 / 50.00



Fig. 6. **Future-video predictions with and without skeleton overlays.** The model without overlays stalls while transporting the wrist guard after grasping it and loses the hand structure needed to grasp the mango under four-finger occlusion in Place Fruit into Box. Full Skel-WAM maintains coherent motion and explicit hand structure in both cases.

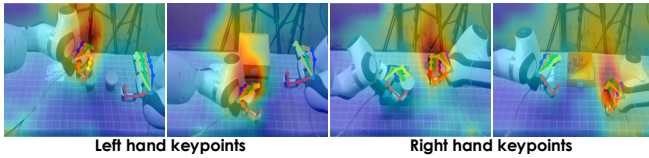


Fig. 7. **Hand-specific keypoint-to-video attention.** Left and right columns show attention from the respective hand keypoint streams, concentrated on corresponding hand regions, indicating spatial alignment with video features.

Fruit into Box, and target drawer height.

Human-robot cotraining raises Skel-WAM’s average SR from 38.89% to 86.11% and PSR from 61.11% to 95.14% (Table II). The gains span all four variations. For drawer height, cotraining increases SR from 0% to 100%, demonstrating that human videos enable Skel-WAM to acquire task cases absent from the robot data. LDA-cotrain directly aligns human wrist poses and MANO parameters with robot end-effector poses and dexterous-hand actions, yet achieves only 30.56% SR and 49.31% PSR. This suggests that directly bridging embodiment-specific pose and action spaces retains a substantial transfer gap. Skel-WAM instead maps human and robot motion into a shared skeleton space, allowing human videos to supervise shared visual and skeletal dynamics without robot action labels.

Ablations confirm the complementary roles of skeleton overlays and structured keypoints. Average SR drops from 86.11% to 50.00% without overlays and below 17% without keypoints, regardless of overlays. Keypoints encode finger-

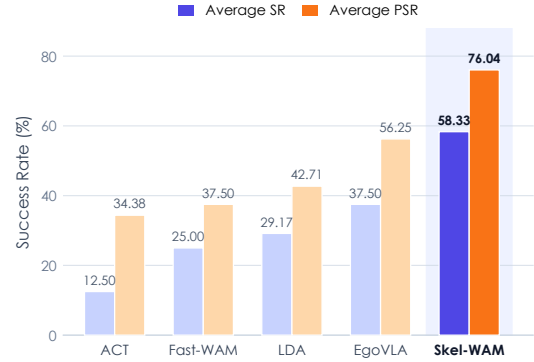


Fig. 8. **Background generalization with robot-only training.** Average SR and PSR across four tasks under an unseen background.

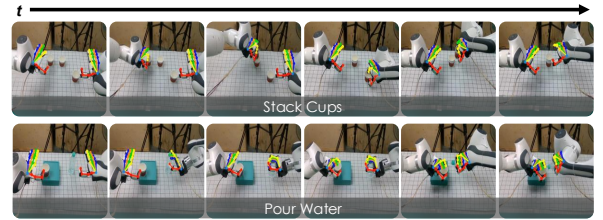


Fig. 9. **Future-video predictions under an unseen background.** For Stack Cups and Pour Water, Skel-WAM maintains coherent task-relevant motion and preserves the test-time background throughout the predicted horizon rather than reverting to the training background shown in the inset.

level trajectories, while overlays ground them in the scene. Full Skel-WAM maintains coherent wrist-guard transport and hand structure under occlusion during mango grasping; removing overlays causes stalled motion and lost hand structure (Fig. 6), supporting the value of both visual and motion alignment for human-to-robot transfer. Attention maps provide further evidence for this alignment (Fig. 7). The left- and right-hand keypoint tokens attend to their corresponding image regions, showing that the Keypoint Expert preserves hand identity and retrieves spatially aligned visual evidence.

D. Background Generalization

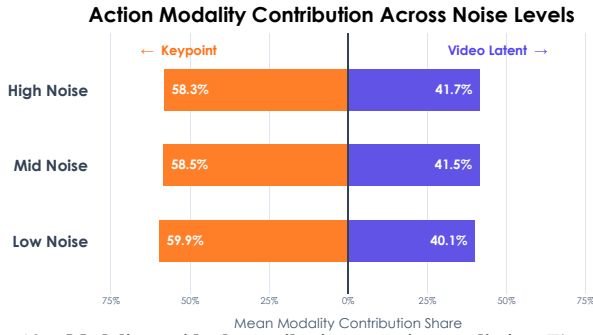
Under the unseen-background setting, Skel-WAM achieves 58.33% average SR and 76.04% PSR, surpassing EgoVLA by 20.83 and 19.79 percentage points, respectively (Fig. 8). These gains show that the learned motion representation remains effective despite changes in scene appearance. Future-video predictions provide complementary evidence (Fig. 9). Skel-WAM maintains coherent task-relevant motion while adapting its visual predictions to the unseen background, as illustrated by the Stack Cups and Pour Water examples.

TABLE III: **Simulation results.** Each entry is SR / PSR (%).

Method	Avg. $\uparrow$	Stack Can	Push Box	Open Drawer	Close Drawer	Flip Mug	Pour Balls	Open Laptop
ACT	24.88 / 55.93	12.90 / 13.98	16.13 / 87.73	22.58 / 70.43	59.14 / 96.77	2.15 / 2.15	5.38 / 64.52	55.91 / 55.91
LDA	34.87 / 61.52	22.58 / 23.66	22.58 / 93.55	21.51 / 76.34	66.67 / 78.49	50.54 / 50.54	2.15 / 43.55	58.06 / 64.52
Fast-WAM	37.33 / 56.22	29.03 / 29.03	19.35 / 89.25	38.71 / 83.33	97.85 / 98.92	24.73 / 26.88	17.20 / 29.57	34.41 / 36.56
EgoVLA	55.01 / 65.31	56.99 / 56.99	64.52 / 80.65	49.46 / 73.16	90.32 / 92.83	4.41 / 4.41	56.99 / 83.56	62.37 / 65.59
Skel-WAM	63.29 / 78.42	58.06 / 59.14	66.67 / 97.85	49.46 / 86.02	100.00 / 100.00	45.16 / 45.16	27.96 / 61.83	95.70 / 98.92

TABLE IV: **Simulation ablation.** Average SR and PSR (%) over seven tasks are reported.

Method	SR	PSR
Skel-WAM	63.29	78.42
w/o keypoints	38.40	58.14
w/o overlay	61.90	77.88
w/o both	34.56	55.53

Fig. 10. **Modality residual contribution to action prediction.** The gated residual RMS share is averaged across action frames at three denoising noise levels. Both conditions contribute throughout denoising, with a consistently larger keypoint contribution.

V. SIMULATION EXPERIMENTS

We evaluate Skel-WAM on seven EgoVLA-Sim [7] tasks to answer two questions: (1) How does the unified skeleton motion interface benefit policy learning from robot demonstrations alone? (2) How do skeleton keypoints and video latents condition robot action prediction?

Each task is evaluated over 93 episodes, including 27 in-distribution episodes and 66 episodes with unseen tables or backgrounds. Object positions are sampled within the benchmark range in both settings. We compare against the same four baselines used in the real-world experiments. For fairness, all models are trained only on robot data, without additional robot pretraining or human-robot cotraining.

Main results and ablation study. As shown in Table III, Skel-WAM achieves 63.29% SR and 78.42% PSR, surpassing EgoVLA by 8.28 and 13.11 percentage points, respectively. The improvement is observed across most evaluated tasks. We attribute these gains primarily to structured keypoint prediction, which provides the Action Expert with explicit future hand geometry and motion cues complementary to the video latents. The following ablations support this interpretation. Table IV evaluates the two skeleton representations. Removing structured keypoints decreases average SR from 63.29% to 38.40%, while removing both representations further reduces it to 34.56%. In contrast, removing only the overlay results in 61.90% SR. Structured keypoints therefore provide most of the robot-only simulation gain. The

limited overlay gain is expected as overlays are most useful during human-robot cotraining, where they reduce the visual appearance gap.

Action modality contribution. To quantify video and keypoint conditioning of the Action Expert, let $\Delta_m \in \mathbb{R}^{d_h}$ denote the residual update from modality $m \in \{V, K\}$ after output projection and, for keypoints, the learned gate. We measure its RMS magnitude as $r_m = \|\Delta_m\|_2 / \sqrt{d_h}$ and its normalized share as $s_m = r_m / (r_V + r_K)$, capturing the update magnitude in the action residual stream rather than raw attention weights. Averaged across action frames, keypoints account for 58.3-59.9% across noise levels and video latents for 40.1-41.7% (Fig. 10). Both modalities contribute to action prediction, with keypoints producing larger residual updates.

VI. CONCLUSIONS

We introduced Skel-WAM, a world action model for manipulation that aligns human and robot hand motion through visual skeleton overlays and structured keypoints. A decoupled Video-Keypoint MoT learns shared visual and skeletal dynamics, while a robot-only causal Action Expert maps them to executable controls, enabling human videos to be used without robot action labels. Skel-WAM consistently outperforms strong baselines across real-world, background-shift, and simulation settings. Human-robot cotraining further raises success from 38.89% to 86.11% on four robot-missing variations under fixed task language. These results show that unified hand-skeleton motion transforms human demonstration diversity into broader coverage of task variations for robot policies.

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